Math Inquirers

Extension Workbook

with answers

Level 3

Self-study | Ages 8-10



My name _____

About this book

Math Inquirers Extension Workbook, Level 3
Self-study practice with answers.

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Good to know

- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
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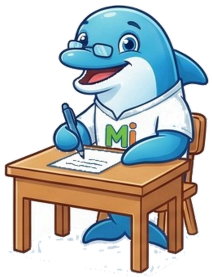
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Welcome!



This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Printing this soft copy? Print at 100 percent so the rulers are true size.

The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	Number sleuths (numbers to 10 000)	Not in this proof edition				☐
2	Cryptarithm crackers	6-7	8-9	10 and 12	11	☐
3	Times table treasure	Not in this proof edition				☐
4	Model makers (bar models)	Not in this proof edition				☐
5	Remainder riddles	Not in this proof edition				☐
6	Fraction walls	Not in this proof edition				☐
7	Area architects	Not in this proof edition				☐
8	Angle and line detectives	Not in this proof edition				☐
9	Measure and money masters	Not in this proof edition				☐
10	Puzzle day	Not in this proof edition				☐
11	Catch-up week	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

Learn it: Addition and subtraction

My goal. I can find missing digits in sums and take-aways. I can solve magic squares and number pyramids.

In a missing-digit sum, work one column at a time. Start with the ones. Remember to add the digit you carried.

Worked example

Find the 2 missing digits in this sum.

	Th	H	T	O
	4	5	□	7
+	1	□	6	8
	5	8	9	5

1. Ones: 7 + 8 = 15. Write 5 and carry 1.
2. Tens: ? + 6 + 1 carried must end in 9. So ? is 2.
3. Hundreds: 5 + ? makes 8. So ? is 3.
4. Check the thousands: 4 + 1 = 5. It matches.

Answer: 4527 + 1368 = 5895. The missing digits are 2 and 3.

How to find a missing digit

1. Look at the ones column first.
2. Write the carried 1 above the next column.
3. For a take-away, turn it round and add to check.

Your turn

Answer at the back of the book.

Try **Find the missing digit.** □

3?48 + 2517 = 5965

Start here

Do these first. Take your time.

1 **Find the missing digit in each sum.**

(a) 3?46 + 1251 = 4597 The missing digit is ☐.

(b) 26?9 + 1435 = 4104 The missing digit is ☐.

2 **Find the missing digit in each take-away.**

(a) 6405 − 2?18 = 3687 The missing digit is ☐.

(b) 9?30 − 4617 = 4513 The missing digit is ☐.

3 **A theatre sells 3875 tickets on Saturday and 2968 tickets on Sunday.**

(a) It sells ☐ tickets in all.

(b) It sells ☐ more tickets on Saturday than on Sunday.

4 **A baker has 8000 g of flour. He uses 2475 g on Monday and 3150 g on Tuesday. How many grams of flour are left?** ☐ g

5 **Siti works out 5000 − 1768 = 3342.**

(a) Check by adding: 3342 + 1768 = ☐

(b) Is Siti right? Yes No

(c) The correct answer is ☐.

6 **Look for a pattern.**

(a) 2641 + ☐ = 9999

(b) 10 000 − 2641 = ☐

Start here

Keep going. Check your work.

7 **Every row, column and diagonal has the same total.**

(a) The magic total is ☐.

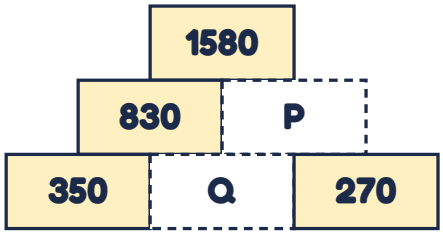
(b) A = ☐ B = ☐ C = ☐

200	450	A
550	350	150
300	B	C

8 **A brick is made by adding the 2 bricks below it.**

(a) P = ☐

(b) Q = ☐



9 **Find the 2 missing digits.**

(a) The digit in the hundreds place of the top number is ☐.

(b) The digit in the tens place of the number being added is ☐.

Th	H	T	O
3	☐	5	2
+ 2	4	☐	6
6	1	4	8

10 **Each letter stands for a digit. The same letter always means the same digit. Find the digit.**

(a) A3 + 2A = 56 A = ☐

(b) C5 + C5 = 90 C = ☐

Start here

Nearly there. You can do it.

11 **Find the 2 missing digits.**

(a) The digit in the hundreds place of the top number is ☐.

(b) The digit in the tens place of the number taken away is ☐.

	Th	H	T	O
	6	☐	0	2
−	2	8	☐	5
	3	6	5	7

12 **The sum of 2 numbers is 9000. One number is 3475. What is the other number?** ☐

Working: _____

13 **Use the cards once each to make two 3-digit numbers.**

2	3	4	5	7	8

(a) The greatest sum is ☐.

(b) The smallest sum is ☐.

14 **A school collects cans for 3 weeks: 1875, 2430 and 1790.**

(a) In all, the school collects ☐ cans.

(b) Week 2 has ☐ more cans than Week 3.

Working: _____

Think harder

Hints are upside down. Try first!

1 **Letters hide digits. The same letter is the same digit. Different letters are different digits.**

AB + AC = BBD

(a) A = ☐ (b) B = ☐ (c) C = ☐ (d) D = ☐

My tries: _____

_____

Hint. Two 2-digit numbers add up to less than 200. What is B?

2 **Use the cards 1, 3, 5, 6 and 8 once each. Make a 3-digit number and a 2-digit number. What is the smallest sum you can make?** ☐

1	3	5	6	8

My tries: _____

_____

_____

Hint. Which digits are worth the least? Which places are worth most?

3 **Look at this magic square. Every row, column and diagonal has the same total.**

(a) The magic total is ☐.

(b) A = ☐

(c) E = ☐

120	270	A
	210	
180		E

Hint. The middle number is in 4 lines. Add those 4 lines.

Go further

A weekend page. Big challenges. Hints are upside down.

1 **Letters hide digits. The same letter is the same digit. Different letters are different digits.**

ABC + BC = CAB

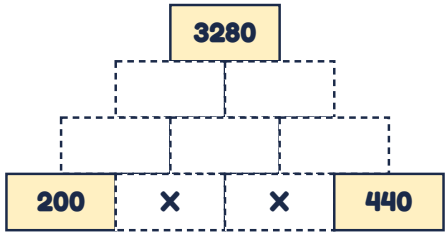
(a) A = ☐ (b) B = ☐ (c) C = ☐

My tries: _____

Hint. Look at the ones: C + C ends in B. List the choices for C.

2 **Beyond Level 3: Pyramids with a hidden number.**

A brick is made by adding the 2 bricks below it. The 2 bricks marked X are equal. Find X. ☐



Hint. Guess a number for X. Fill the pyramid. Is the top too big?

3 **Beyond Level 3: Reversing digits.**

Reverse the digits of a 2-digit number and add the 2 numbers.

(a) 36 + 63 = ☐ (b) 58 + 85 = ☐ (c) 47 + 74 = ☐

(d) Each answer is in the ☐ times table.

Hint. Add the tens digit and ones digit. Compare with the answers.

4 **Make up a number pyramid puzzle of your own. The bottom row has 3 bricks. Show exactly 3 bricks and hide the other 3, so that only ONE answer fits. Swap puzzles with a grown-up.**

Draw my pyramid:

Hint. Make a full pyramid first. Then hide 3 bricks.

✔ Check yourself

No hints. Try each one on your own.

1 **Find the missing digit.** ☐

4?72 + 2519 = 7091

Working: _____

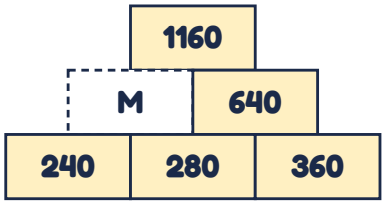
2 **Find the missing digit.** ☐

8?04 − 3569 = 4635

Working: _____

3 **A brick is made by adding the 2 bricks below it.**

The brick M is ☐ **.**



4 **A zoo has 9000 visitors in 3 days. On Friday 2845 visitors come. On Saturday 3260 come. How many come on Sunday?** ☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Number sleuths (numbers to 10 000)	☐	☐	☐	☐	/ 4	
2	Cryptarithm crackers	☐	☐	☐	☐	/ 4	
3	Times table treasure	☐	☐	☐	☐	/ 4	
4	Model makers (bar models)	☐	☐	☐	☐	/ 4	
5	Remainder riddles	☐	☐	☐	☐	/ 4	
6	Fraction walls	☐	☐	☐	☐	/ 4	
7	Area architects	☐	☐	☐	☐	/ 4	
8	Angle and line detectives	☐	☐	☐	☐	/ 4	
9	Measure and money masters	☐	☐	☐	☐	/ 4	
10	Puzzle day	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 3.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 2 · Cryptarithm crackers

Your turn
page 6

Answer: 4
Why: Ones: 8 + 7 = 15, write 5 and carry 1. Tens: 4 + 1 + 1 = 6. Hundreds: ? + 5 makes 9, so ? is 4. Thousands: 3 + 2 = 5.

START HERE

1
page 7

Answer: (a) 3 (b) 6
Why: (a) Ones 6 + 1 = 7 and tens 4 + 5 = 9 are fine. Hundreds: ? + 2 = 5, so ? is 3. (b) Ones: 9 + 5 = 14, carry 1. Tens: ? + 3 + 1 ends in 0, so ? is 6 and 10 is made. Check: 2669 + 1435 = 4104.

2
page 7

Answer: (a) 7 (b) 1
Why: Turn each one round and add. (a) 3687 + 2718 = 6405, so the digit is 7. (b) 4513 + 4617 = 9130, so the digit is 1.

3
page 7

Answer: (a) 6843 (b) 907
Why: (a) 3875 + 2968 = 6843. (b) 3875 - 2968 = 907. Check: 2968 + 907 = 3875.

4
page 7

Answer: 2375
Why: First add what he used: 2475 + 3150 = 5625. Then 8000 - 5625 = 2375. Check: 5625 + 2375 = 8000.

5
page 7

Answer: (a) 5110 (b) No (c) 3232
Why: The check should give back 5000, but 3342 + 1768 = 5110. So Siti is not right. 5000 - 1768 = 3232. Check: 3232 + 1768 = 5000.

6
page 7

Answer: (a) 7358 (b) 7359
Why: (a) Each digit and its partner add up to 9: 2 and 7, 6 and 3, 4 and 5, 1 and 8. That gives 7358. (b) 10 000 is 1 more than 9999, so the answer is 1 more than 7358.

7
page 8

Answer: (a) 1050 (b) A = 400, B = 250, C = 500
Why: The middle row is complete: 550 + 350 + 150 = 1050. Top row: 200 + 450 + A = 1050, so A = 400. Left column: 200 + 550 + 300 = 1050. Middle column: 450 + 350 + B = 1050, so B = 250. Bottom row: 300 + 250 + C = 1050, so C = 500.

8
page 8

Answer: (a) 750 (b) 480
Why: P and 830 make the top brick: 830 + P = 1580, so P = 750. Q and 270 make P: Q + 270 = 750, so Q = 480. Check: 350 + 480 = 830.

9
page 8

Answer: (a) 6 (b) 9
Why: Ones: 2 + 6 = 8. Tens: 5 + ? ends in 4, so ? is 9 and 1 is carried. Hundreds: ? + 4 + 1 ends in 1, so ? is 6 and 1 is carried. Thousands: 3 + 2 + 1 = 6. Check: 3652 + 2496 = 6148.

10
page 8

Answer: (a) 3 (b) 4
Why: (a) Try A = 3: 33 + 23 = 56. It works. (b) Two equal numbers add up to 90, so each is 45. Then C = 4.

11 page 9

Answer: (a) 5 (b) 4
Why: Turn it round: 3657 + 2845 = 6502 (the missing digit of the number taken away is its tens digit 4). Or subtract by columns: 12 - 5 = 7, then 10 - 1 - 4 = 5, then 15 - 1 - 8 = 6, then 6 - 1 - 2 = 3. The top number is 6502.

12 page 9

Answer: 5525
Why: 9000 - 3475 = 5525. Check: 3475 + 5525 = 9000.

13 page 9

Answer: (a) 1595 (b) 605
Why: For the greatest sum, put the 2 greatest digits in the hundreds places. Then the next 2 in the tens places, and the last 2 in the ones places: 800 + 700 + 50 + 40 + 3 + 2 = 1595. For the smallest sum, do the same with the smallest digits first: 200 + 300 + 40 + 50 + 7 + 8 = 605.

14 page 9

Answer: (a) 6095 (b) 640
Why: (a) 1875 + 2430 = 4305, and 4305 + 1790 = 6095. (b) 2430 - 1790 = 640. Check: 1790 + 640 = 2430.

THINK HARDER

1 page 10

Answer: (a) 5 (b) 1 (c) 9 (d) 0
Why: Two 2-digit numbers make less than 200, so B = 1 and the answer starts 11. Ones: B + C = 1 + C ends in D. Tens: A + A + carry makes 11, so A = 5 with carry 1. Then 1 + C must be 10, so C = 9 and D = 0. Check: 51 + 59 = 110.

2 page 10

Answer: 194
Why: The smallest digit, 1, goes in the hundreds place. The next smallest, 3 and 5, go in the two tens places. The others, 6 and 8, go in the ones places. For example 136 + 58 = 194, and every way of placing the digits like this gives 194.

3 page 10

Answer: (a) 630 (b) 240 (c) 300
Why: Add the middle row, the middle column and the 2 diagonals. The middle number is counted 4 times, and every other number once, so 4 totals = all 9 numbers + 3 times the middle number. All 9 numbers make 3 totals, so the total is 3 times the middle: 3 x 210 = 630. Then A = 630 - 120 - 270 = 240 and E = 630 - 120 - 210 = 300.

GO FURTHER

1 page 11

Answer: (a) 2 (b) 6 (c) 3
Why: Ones: C + C ends in B. Tens: B + B (+ carry) ends in A. Hundreds: A (+ carry) is C. Try C = 3: then B = 6, carry 0. Tens: 6 + 6 = 12, so A = 2 and carry 1. Hundreds: 2 + 1 = 3 = C. Check: 263 + 63 = 326. No other digits work.

2 page 11

Answer: 440
Why: In this pyramid the end bricks of the bottom row count once in the top brick and the 2 middle bricks count 3 times each. So 200 + 3 X + 3 X + 440 = 3280. That is 640 + 6 X = 3280, so 6 X = 2640 and X = 440. Check: the next row is 640, 880, 880; then 1520, 1760; then 3280.

3
page 11

Answer: (a) 99 (b) 143 (c) 121 (d) 11
Why: 99 = 9 x 11, 143 = 13 x 11, 121 = 11 x 11. Here is why. A 2-digit number with digits A and B is 10 A + B. Its reverse is 10 B + A. Together they make 11 A + 11 B, which is 11 x (A + B). So the answer is always in the 11 times table.

4
page 11

Answer: Answers vary. Example: top brick 800, middle left brick 370, bottom right brick 180 shown. The hidden bricks are 430 (middle right), 250 (bottom middle) and 120 (bottom left).
Why: Check that your pyramid adds up correctly and that the 3 shown bricks fix all the others. In the example: 800 - 370 = 430, then 430 - 180 = 250, then 370 - 250 = 120.

CHECK YOURSELF

1
page 12

Answer: 5
Why: Ones: 2 + 9 = 11, carry 1. Tens: 7 + 1 + 1 = 9. Hundreds: ? + 5 ends in 0, so ? is 5 and 1 is carried. Thousands: 4 + 2 + 1 = 7.

2
page 12

Answer: 2
Why: Turn it round: 4635 + 3569 = 8204, so the missing digit is 2.

3
page 12

Answer: 520
Why: M is the sum of 240 and 280: 240 + 280 = 520. Check: 520 + 640 = 1160.

4
page 12

Answer: 2895
Why: 2845 + 3260 = 6105. Then 9000 - 6105 = 2895. Check: 6105 + 2895 = 9000.